Chapter 19

Applications Of Inversive Geometry

Peaucellier-Lipkin Linkage

In order to get the formalities out of the way as early as possible, we remind our readers about the following historical events. In a private 1864 letter the officer of the Engineering Arm of the French Army Charles-Nicolas Peaucellier (1832–1913) stated that he invented a linkage that now carries his name, a planar mechanism that can convert circular motion into its linear equivalent and conversely with mathematical precision*.

Apparently, however, officer Peaucellier did not supply any exact details on how the said linkage works.

In 1868 the student of the Russian mathematician P. L. Chebyshev (1821–1894) by the name Lipman Lipkin (1840–1876) invented a similar linkage independently. The detailed paper by Mr. Lipkin was published in 1870. In 1873 Mr. Peaucellier published his paper with the reference to the paper by Mr. Lipkin.

Be that as it may, the linkage that we will discuss in this chapter carries the names of both inventors and is now known as *the Peaucellier-Lipkin linkage*.

Since around the time when [James Watt (1736–1819)] invented a better steam engine, among the engineers there was an interest in developing a mechanism that could convert a circular motion of one linkage joint into a linear motion of another linkage joint. Such a mechanism, as most mechanical devices go, should be rugged and robust, simple in its correct operation and maintenance and, ideally, such a mechanism should be compact or *planar*.

The devices that converted the circular motion into its linear counterpart did exist at the time but the degree of the circular-to-linear conversion that these devices provided was only approximate and, further, the conversion itself only applied to small amplitudes of motion.

Apparently, the Peaucellier-Lipkin linkage was the first device that provided a planar solution of the problem which was mathematically precise. As we shall see momentarily, there is nothing approximate in the operation of the said linkage as the mathematics on which the device is based is ironclad solid.

*Having said that, we hope that our readers understand very well that any switch from the ideal world of mathematics to the world of physical reality necessarily introduces approximations and, hence, errors. In real life there is no such thing as a mathematical *point*, *circle*, *straight line* or *plane*. Such mathematical objects play the roles of **models** based on which the various real life implementations are constructed.

As such, when we say that the Peaucellier-Lipkin linkage is *planar*, we, of course, do not mean that the said device actually fits into a mathematical plane. What we mean here is that the relevant **centers of mass**, which can be modeled with mathematical points unreasonably well, execute a motion in a real space that can be modeled with a mathematical surface known as *a plane* unreasonably well.

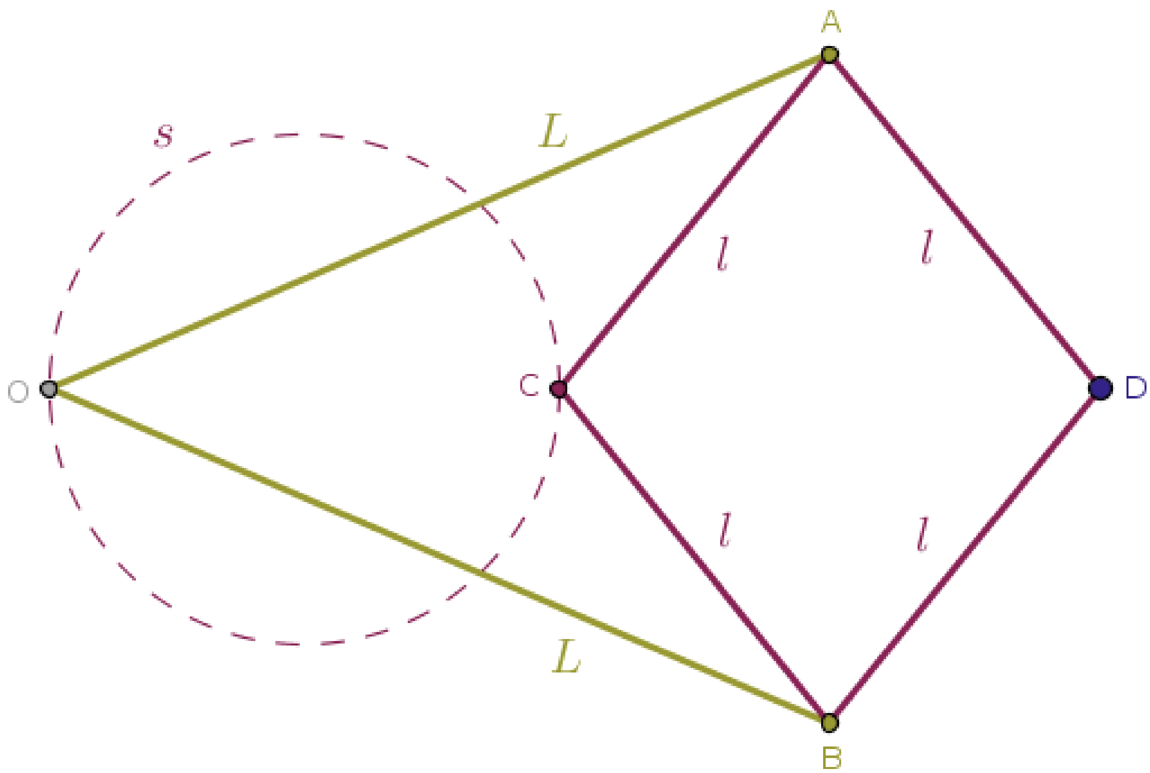
Moreover, while a physical linkage joint in real life is not a mathematical point, its center of mass can be modeled with one unreasonably well. Thus, when in the upcoming discussion we will state that a joint so and so *is moving in a circle*, it really means that the center of mass of a given joint executes a motion in a real space that can be modeled with a mathematical curve known as *a circle* unreasonably well and so on.

Cleaving all the irrelevant details with the razor of Occam away, we now understand that fundamentally the Peaucellier-Lipkin linkage consists of just six rigid rods, each of which is modeled with a mathematical line segment:

- the two rigid rods OA and OB of equal length L and
- the four rigid rods AC, AD, BC and BD of equal length $l < L$

The two rods of equal length L are linked and joined to a fixed point, O, through one of the extremities of each rod.

The four rods of equal length l are linked and joined into a rhombus $ADBC$ trapped between the two rods of length L (Figure 19.1):



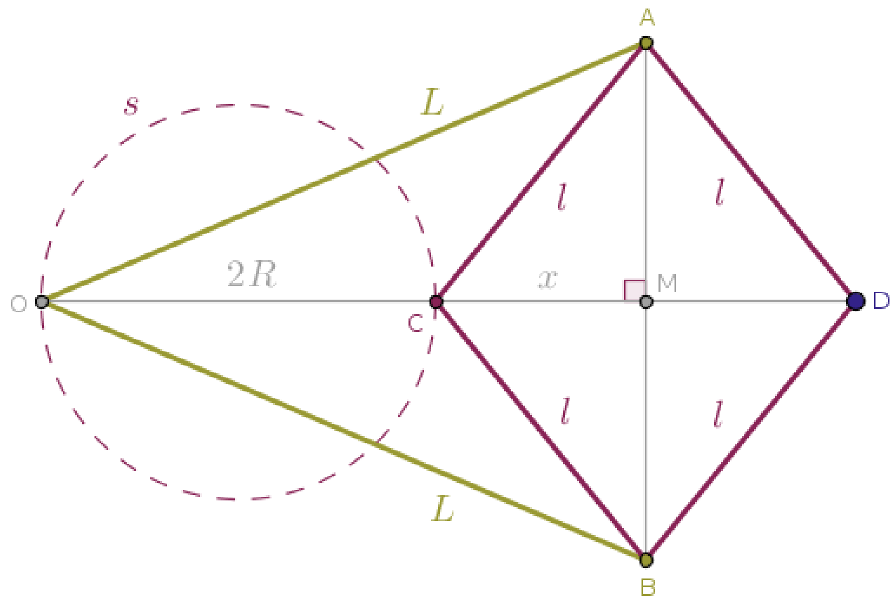
All six rods are linked into the joints that are free to move with respect to each other. In a traditional or classical Peaucellier-Lipkin linkage the input power is applied to the joint C in such a way that the point C traces an arc of a parent circle, s, with the radius R that passes through the said fixed point, O.

Using the configuration of mathematical objects shown in the Figure 19.1 as an opportunity to generate yet another proof, since a student of mathematics will never miss such a chance, we put together the following

Problem 25. Given the classical Peaucellier-Lipkin linkage shown in Figure 19.1 above, what curve will the point D trace?

Again, in this problem we take it that the point C traces an arc of the circle, s, that passes through the fixed point O.

Solution. In order to answer the above question we, pretending that we know absolutely nothing about inversion, first catch the point C in its motion in a position when the said point belongs to the other extremity of a diameter of the circle s that also passes through the point O (Figure 19.2):



Let M be the point where the diagonals AB and CD of the rhombus $ADBC$ intersect and let $x = |MC| = |MD|$. Then, from the right triangle OMA we have:

$$|OA|^2 = |OM|^2 + |AM|^2$$

which, via the right triangle CMA, is to say that:

$$L^2 = (2R + x)^2 + l^2 - x^2$$

which is, further, to say that:

$$L^2 = 4R^2 + 4Rx + l^2$$

Solving the above equation for x, we find:

$$x = \frac{L^2 - l^2}{4R} - R \qquad (1)$$

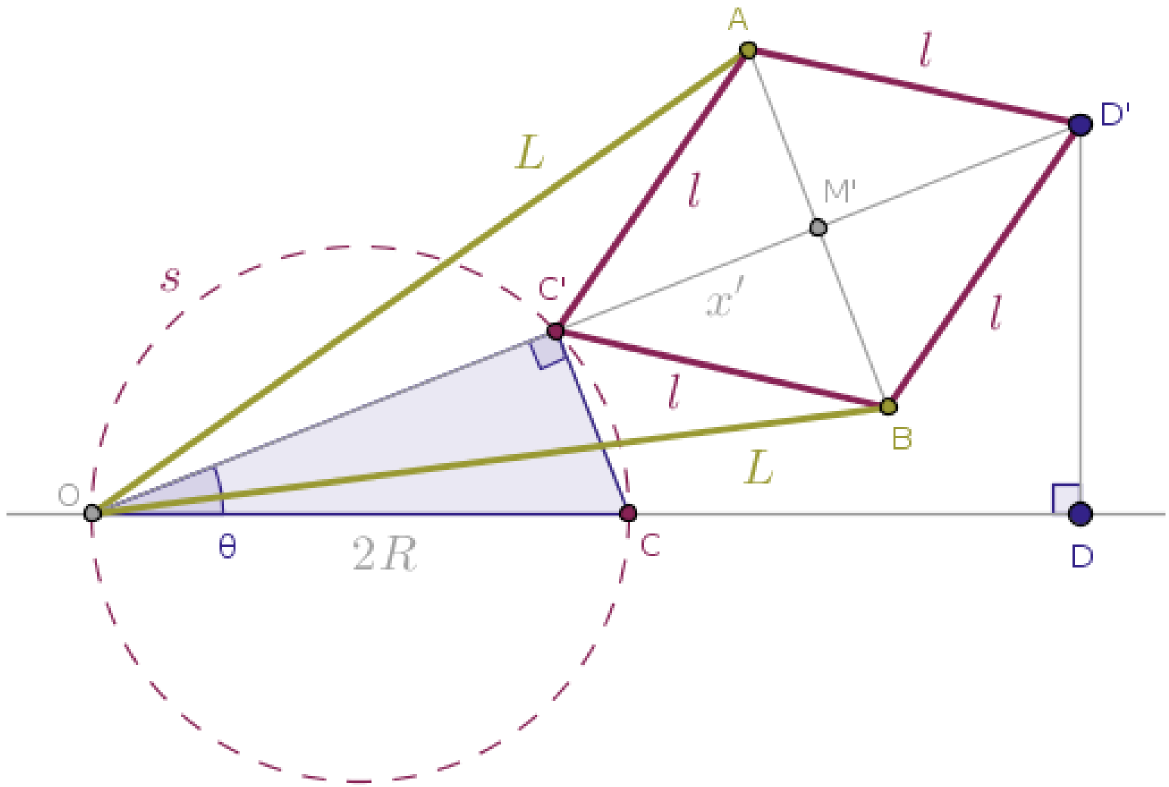
Correspondingly, for the position of the point D with respect to the fixed point O in that state of the linkage, we have:

$$|OD| = |OC| + |CD| = 2R + 2x$$

which is to say that with (**1**):

$$|OD| = \frac{L^2 - l^2}{2R} \tag{2}$$

Next, we catch the point C in its motion when it turned by an arbitrary but legal angle θ about the fixed point O (Figure 19.3):



Because the type of motion that our linkage executes is rigid, while the mutual juxtaposition of the elements of that linkage will change, the lengths of the rods that make up the said linkage will not.

Thus, the shape $AD'BC'$ will still be ***a rhombus***, a rhombus that is different from its counterpart shown in Figure 19.2, but it will be a rhombus nonetheless. Hence, the diagonals AB and $C'D'$ of the rhombus $AD'BC'$ will intersect at the point M' at right angles still.

Therefore, from the right triangles $OM'A$ and $C'M'A$, we shall have:

$$L^2 = (|OC'| + x')^2 + l^2 - x'^2$$

The interior angle at the vertex C' is right due to Euclid's "Elements" Book 3 Proposition 31, also known as the Thales's Theorem, since that angle $OC'C$ subtends the diameter OC of the parent circle s.

Hence:

$$|OC'| = |OC| \cdot \cos(\theta) = 2R\cos(\theta)$$

Therefore:

$$L^2 = (2R\cos(\theta) + x')^2 + l^2 - x'^2$$

Solving the above equation for x', we find:

$$x' = \frac{L^2 - l^2}{4R\cos(\theta)} - R\cos(\theta) \tag{3}$$

Correspondingly, for the radial position of the point D' with respect to the fixed point O in that state of the linkage, we have:

$$|OD'| = |OC'| + |C'D'| = 2R\cos(\theta) + 2x'$$

which is to say that with (**3**):

$$|OD'| = \frac{L^2 - l^2}{2R\cos(\theta)} \tag{4}$$

However, the above relation can be rewritten as follows:

$$|OD'|\cos(\theta) = \frac{L^2 - l^2}{2R} \tag{5}$$

Since the RHS of the relation (**5**) matches the RHS of the relation (2), we conclude that:

$$|OD'|\cos(\theta) = |OD| \tag{6}$$

and the above relation tells us that the orthogonal projection of the point D' onto the straight line OC extended coincides with the location of the point D. Since the angle θ and, hence, point C' was picked at random, it follows that as the point C traverses an arc of the circle, s, the point D travels along *a straight line*, which is what was required to prove. □

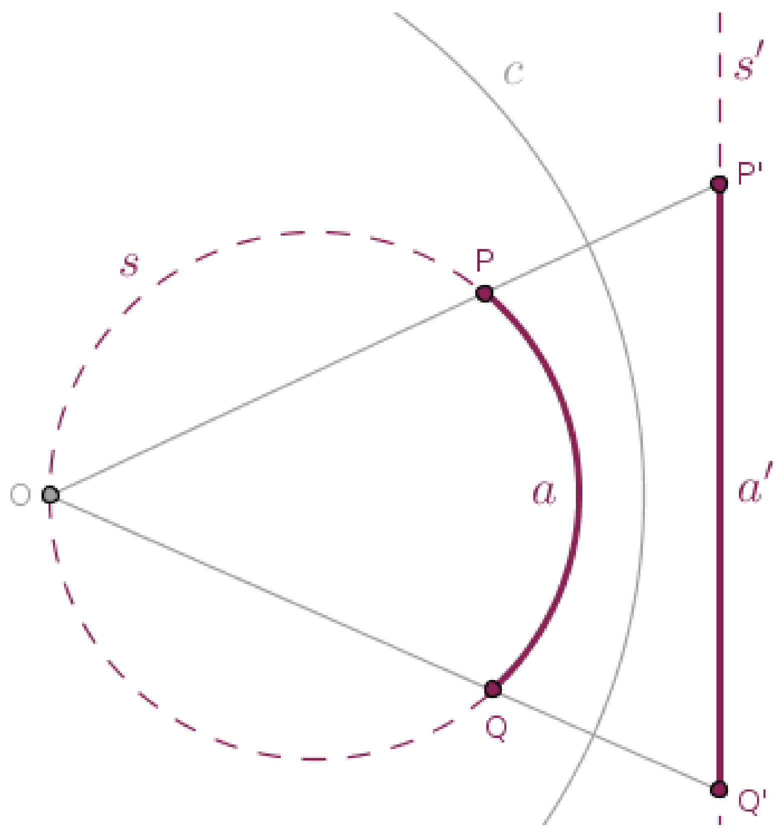
Now, as the above upper middle school or lower high school deduction shows, it is entirely possible that, in theory, a person with just some basic knowledge of the elementary Euclidean geometry can stumble upon the arrangement of the line segments shown across in the figures above and reach a conclusion shown in the relation (**6**) by some miraculous sequence of fortunate events.

However, such a person will inevitably ask the following question: hm, is this just a flaky coincidence or is there some deeper mathematics behind it?

Well. As we stop pretending that we know nothing about inversive geometry, we can now answer that question in the positive: yes, there is some deeper mathematics behind the Peaucellier-Lipkin linkage and that is the mathematics of inversive geometry.

Due to the cluster of the theorems numbered 15 through 19, we know that under any inversion of the real plane with respect to a circle c centered at the point O, any circle that passes through the center of inversion, O, will be reflected in c into a straight line that does not pass through the said center of inversion regardless of the power of inversion, positive or negative.

Therefore, under an inversion with respect to a circle $c(O, r)$ an image of an arc a of the parent circle s that passes through the point O will be the corresponding line segment a' of the parent straight line s' into which the circle s is reflected in c with a given power (Figure 19.4):



Above, we have shown a sample inversion with respect to a circle with positive power and we, of course, assume that the circular arc a does not pass through the center of inversion, O.

We now quickly prove that due to the symmetry of the linkage, the points O, C, D remain collinear at all times and so are the points O, C', D'.

If we construct a product $|OC| \cdot |OD|$ corresponding to the state of the linkage captured in Figure 19.2 then we shall see that it comes out to be a constant real number:

$$|OC| \cdot |OD| = 2R \cdot \frac{L^2 - l^2}{2R} = L^2 - l^2 = \text{const}$$

If we construct a product $|OC'| \cdot |OD'|$ corresponding to the state of the linkage captured in Figure 19.3 then we shall see that it comes out to be *the same* constant real number:

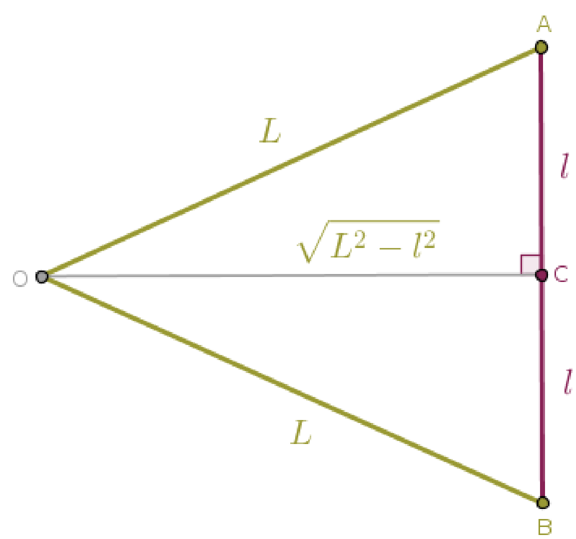
$$|OC'| \cdot |OD'| = 2R\cos(\theta) \cdot \frac{L^2 - l^2}{2R\cos(\theta)} = L^2 - l^2 = \text{const}$$

and we already understand the meaning of such a constant real number. That number represents the square of the length of the radius of inversion, r:

$$L^2 - l^2 = r^2 \tag{7}$$

and, technically, *the power* of the said inversion, positive, since $L > l$.

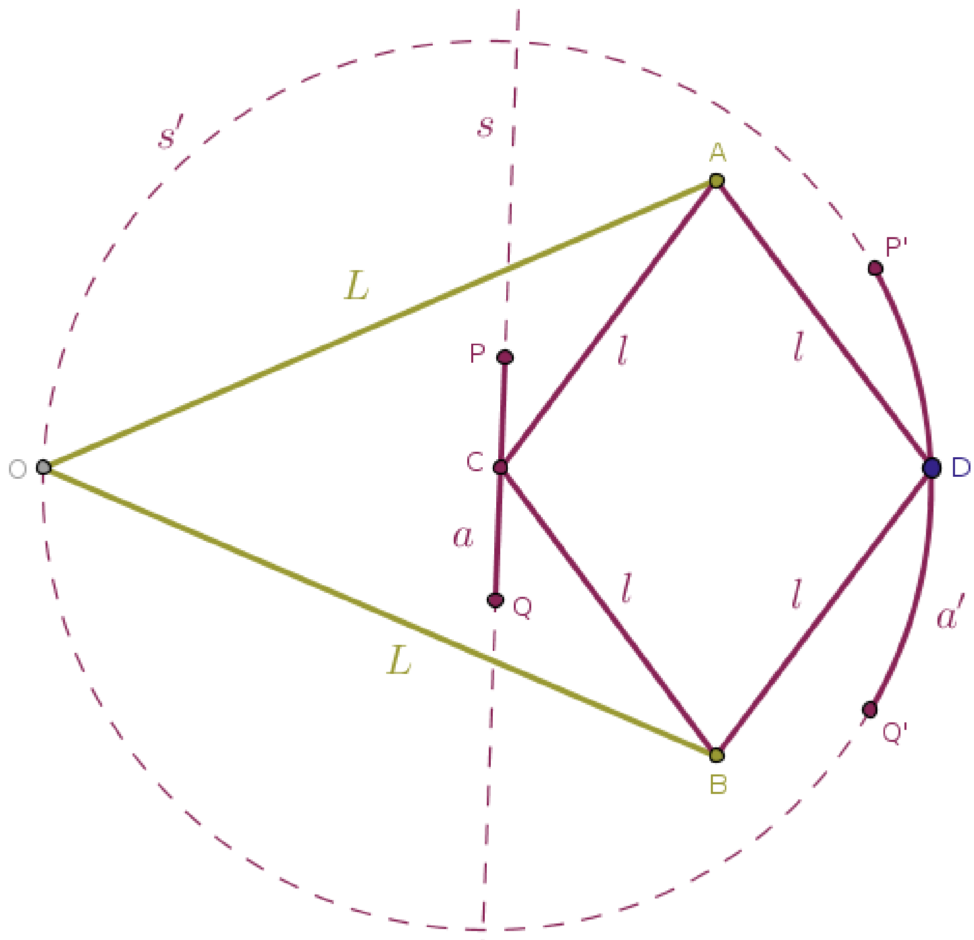
We note in passing that the geometric interpretation of the relation in (**7**) is straightforward. When all three points C, M and D collapse into a single point, C, then the linkage degenerates into an isosceles triangle (Figure 19.5):



Thus, the Peaucellier-Lipkin linkage represents a physical implementation of a mathematically abstract concept of inversion for under the inversion or the parent plane with positive power with respect to a circle, c, centered at the fixed point O with the radius, r, whose magnitude is shown in (**7**), the above points C and D are the inverses of each other.

But, in conclusion, we can say even more:

- if the input power is applied to the point D in such a way that the point D traces a line segment a' shown in Figure 19.4 then the point C will trace the corresponding arc, a, that belongs to the parent circle, s, that passes through the center of inversion, O
- if the input power is applied to the point C in such a way that the point C runs a long a line segment, a, that belongs to the parent straight line, s, that does not pass through the center of inversion, O, then the point D will trace a circular arc, a', that belongs to the corresponding parent circle, s', that passes through the center of inversion (Figure 19.6):



- and conversely, if the input power is applied to the point D in such a way that the point D traces an arc, a', of a circle, s', that passes through the center of inversion, O, then the point C will trace a line segment, a, that belongs to the corresponding parent straight line, s, that does not pass through the center of inversion

and so on.

The mathematical heft of the above observations boils down to the following topological truth:

topologically, as we have proven in our previous theorems, circles and straight lines are **equivalent** in the following specific way, any inversion of a plane with respect to a circle transforms straight lines that do not pass through the center of inversion into circles that do pass through the center of inversion and conversely; any such inversion transforms circles that pass through the center of inversion into straight lines that do not pass through the center of inversion

We hope that even after our short analysis of the inner workings of the Peaucellier-Lipkin linkage, our readers are beginning to see that the said linkage is just as rugged, simple and beautiful as the mathematics that it is based on. □